```
MariaDB [employee]> SELECT * FROM EMPLOYEES;
+----+---------+----------+---------------+------------+
| id | name    | salary   | department_id | manager_id |
+----+---------+----------+---------------+------------+
|  1 | Alice   | 60000.00 |             2 |         10 |
|  2 | Bob     | 55000.00 |             2 |         10 |
|  3 | Charlie | 50000.00 |             1 |       NULL |
|  4 | David   | 75000.00 |             4 |          5 |
|  5 | Emma    | 90000.00 |             4 |       NULL |
|  6 | Frank   | 47000.00 |             1 |          3 |
|  7 | Grace   | 52000.00 |             3 |       NULL |
|  8 | Hank    | 80000.00 |             5 |       NULL |
|  9 | Ivy     | 67000.00 |             5 |          8 |
| 10 | Jack    | 72000.00 |             2 |       NULL |
| 11 | Karen   | 62000.00 |             6 |         12 |
| 12 | Laura   | 85000.00 |             6 |       NULL |
| 13 | Mike    | 48000.00 |             7 |         14 |
| 14 | Nina    | 78000.00 |             7 |       NULL |
| 15 | Olivia  | 51000.00 |             3 |          7 |
| 16 | Paul    | 53000.00 |             1 |          3 |
| 17 | Quincy  | 59000.00 |             2 |         10 |
| 18 | Rachel  | 61000.00 |             4 |          5 |
| 19 | Steve   | 64000.00 |          NULL |       NULL |
| 20 | Tracy   | 66000.00 |             6 |         12 |
| 21 | Uma     | 68000.00 |             7 |         14 |
| 22 | Victor  | 70000.00 |             3 |          7 |
| 23 | Wendy   | 72000.00 |             1 |          3 |
| 24 | Xander  | 74000.00 |             2 |         10 |
| 25 | Yvonne  | 70000.00 |          NULL |       NULL |
| 26 | Zach    | 78000.00 |             5 |          8 |
+----+---------+----------+---------------+------------+
26 rows in set (0.001 sec)
```

```
MariaDB [employee]> SELECT * FROM DEPARTMENTS;
+----+------------------+------------+
| id | department_name  | manager_id |
+----+------------------+------------+
|  1 | HR               |          3 |
|  2 | Engineering      |         10 |
|  3 | Marketing        |          7 |
|  4 | Finance          |          5 |
|  5 | Sales            |          8 |
|  6 | IT               |         12 |
|  7 | Customer Support |         14 |
|  8 | IT SECURITY      |       NULL |
|  9 | QA               |       NULL |
+----+------------------+------------+
9 rows in set (0.001 sec)
```

```
MariaDB [(none)]> source C:/Users/abhij/Downloads/EmployeesDatabase.sql;
Query OK, 0 rows affected, 1 warning (0.001 sec)

Database changed
--------------
CREATE TABLE departments (
    id INT PRIMARY KEY,
    department_name VARCHAR(100),
    manager_id INT
)
--------------

ERROR 1050 (42S01) at line 4 in file: 'C:\Users\abhij\Downloads\EmployeesDatabase.sql': Table 'departments' already exists
Query OK, 0 rows affected (0.013 sec)

Query OK, 9 rows affected (0.009 sec)
Records: 9  Duplicates: 0  Warnings: 0

Query OK, 7 rows affected (0.003 sec)
Records: 7  Duplicates: 0  Warnings: 0

Query OK, 19 rows affected (0.002 sec)
Records: 19  Duplicates: 0  Warnings: 0
```

```
MariaDB [employee]> select employees.id, employees.name, departments.department_name from employees inner join departments on emplo
yees.department_id = departments.id;
+----+---------+-----------------+
| id | name    | department_name |
+----+---------+-----------------+
|  1 | Alice   | Engineering     |
|  2 | Bob     | Engineering     |
|  3 | Charlie | HR              |
|  4 | David   | Finance         |
|  5 | Emma    | Finance         |
|  6 | Frank   | HR              |
|  7 | Grace   | Marketing       |
|  8 | Hank    | Sales           |
|  9 | Ivy     | Sales           |
| 10 | Jack    | Engineering     |
| 11 | Karen   | IT              |
| 12 | Laura   | IT              |
| 13 | Mike    | Customer Support |
| 14 | Nina    | Customer Support |
| 15 | Olivia  | Marketing       |
| 16 | Paul    | HR              |
| 17 | Quincy  | Engineering     |
| 18 | Rachel  | Finance         |
| 20 | Tracy   | IT              |
| 21 | Uma     | Customer Support |
| 22 | Victor  | Marketing       |
| 23 | Wendy   | HR              |
| 24 | Xander  | Engineering     |
| 26 | Zach    | Sales           |
+----+---------+-----------------+
24 rows in set (0.010 sec)
```

```
MariaDB [employee]> select e.id, e.name, d.department_name from employees e left join departments d on e.department_id = d.id;
+----+---------+------------------+
| id | name    | department_name  |
+----+---------+------------------+
|  1 | Alice   | Engineering      |
|  2 | Bob     | Engineering      |
|  3 | Charlie | HR               |
|  4 | David   | Finance          |
|  5 | Emma    | Finance          |
|  6 | Frank   | HR               |
|  7 | Grace   | Marketing        |
|  8 | Hank    | Sales            |
|  9 | Ivy     | Sales            |
| 10 | Jack    | Engineering      |
| 11 | Karen   | IT               |
| 12 | Laura   | IT               |
| 13 | Mike    | Customer Support |
| 14 | Nina    | Customer Support |
| 15 | Olivia  | Marketing        |
| 16 | Paul    | HR               |
| 17 | Quincy  | Engineering      |
| 18 | Rachel  | Finance          |
| 19 | Steve   | NULL             |
| 20 | Tracy   | IT               |
| 21 | Uma     | Customer Support |
| 22 | Victor  | Marketing        |
| 23 | Wendy   | HR               |
| 24 | Xander  | Engineering      |
| 25 | Yvonne  | NULL             |
| 26 | Zach    | Sales            |
+----+---------+------------------+
26 rows in set (0.001 sec)
```

```
MariaDB [employee]> select d.id, d.department_name, e.name from employees e right join departments d on e.department_id = d.id;
+----+------------------+---------+
| id | department_name  | name    |
+----+------------------+---------+
|  1 | HR               | Charlie |
|  1 | HR               | Frank   |
|  1 | HR               | Paul    |
|  1 | HR               | Wendy   |
|  2 | Engineering      | Alice   |
|  2 | Engineering      | Bob     |
|  2 | Engineering      | Jack    |
|  2 | Engineering      | Quincy  |
|  2 | Engineering      | Xander  |
|  3 | Marketing        | Grace   |
|  3 | Marketing        | Olivia  |
|  3 | Marketing        | Victor  |
|  4 | Finance          | David   |
|  4 | Finance          | Emma    |
|  4 | Finance          | Rachel  |
|  5 | Sales            | Hank    |
|  5 | Sales            | Ivy     |
|  5 | Sales            | Zach    |
|  6 | IT               | Karen   |
|  6 | IT               | Laura   |
|  6 | IT               | Tracy   |
|  7 | Customer Support | Mike    |
|  7 | Customer Support | Nina    |
|  7 | Customer Support | Uma     |
|  8 | IT SECURITY      | NULL    |
|  9 | QA               | NULL    |
+----+------------------+---------+
26 rows in set (0.001 sec)
```

```
MariaDB [employee]> select e.id as emp_id, e.name as employee, m.id as mgr_id, m.name as manager from employees e join employees m
on e.manager_id = m.id;
+--------+----------+--------+---------+
| emp_id | employee | mgr_id | manager |
+--------+----------+--------+---------+
|      1 | Alice    |     10 | Jack    |
|      2 | Bob      |     10 | Jack    |
|      4 | David    |      5 | Emma    |
|      6 | Frank    |      3 | Charlie |
|      9 | Ivy      |      8 | Hank    |
|     11 | Karen    |     12 | Laura   |
|     13 | Mike     |     14 | Nina    |
|     15 | Olivia   |      7 | Grace   |
|     16 | Paul     |      3 | Charlie |
|     17 | Quincy   |     10 | Jack    |
|     18 | Rachel   |      5 | Emma    |
|     20 | Tracy    |     12 | Laura   |
|     21 | Uma      |     14 | Nina    |
|     22 | Victor   |      7 | Grace   |
|     23 | Wendy    |      3 | Charlie |
|     24 | Xander   |     10 | Jack    |
|     26 | Zach     |      8 | Hank    |
+--------+----------+--------+---------+
17 rows in set (0.010 sec)
```

```
MariaDB [employee]> select e.id, e.name, d.department_name from employees e join departments d on e.department_id = d.id;
+----+---------+------------------+
| id | name    | department_name  |
+----+---------+------------------+
|  1 | Alice   | Engineering      |
|  2 | Bob     | Engineering      |
|  3 | Charlie | HR               |
|  4 | David   | Finance          |
|  5 | Emma    | Finance          |
|  6 | Frank   | HR               |
|  7 | Grace   | Marketing        |
|  8 | Hank    | Sales            |
|  9 | Ivy     | Sales            |
| 10 | Jack    | Engineering      |
| 11 | Karen   | IT               |
| 12 | Laura   | IT               |
| 13 | Mike    | Customer Support |
| 14 | Nina    | Customer Support |
| 15 | Olivia  | Marketing        |
| 16 | Paul    | HR               |
| 17 | Quincy  | Engineering      |
| 18 | Rachel  | Finance          |
| 20 | Tracy   | IT               |
| 21 | Uma     | Customer Support |
| 22 | Victor  | Marketing        |
| 23 | Wendy   | HR               |
| 24 | Xander  | Engineering      |
| 26 | Zach    | Sales            |
+----+---------+------------------+
24 rows in set (0.001 sec)
```

```
MariaDB [employee]> select id, name, salary from employees where salary in (select max(salary) from employees);
+----+------+----------+
| id | name | salary   |
+----+------+----------+
|  5 | Emma | 90000.00 |
+----+------+----------+
1 row in set (0.001 sec)

MariaDB [employee]> select e.id, e.name, d.department_name, e.salary from employees e join departments d on e.department_id = d.id
where e.salary in (select max(salary) from employees);
+----+------+-----------------+----------+
| id | name | department_name | salary   |
+----+------+-----------------+----------+
|  5 | Emma | Finance         | 90000.00 |
+----+------+-----------------+----------+
1 row in set (0.011 sec)

MariaDB [employee]>
```